Grouped frequency tables and histo[redacted]ns

1 For the following surveys, state whether the data found should be grouped or ungrouped when constructing a frequency table:

a A survey conducted of 1000 people, asking them how many languages they speak.

b A survey conducted of 1000 people, asking them how many different countries they know the names of.

2 Find the class centre of the following class intervals:

a $27 - 40$

b $19 \leq t < 23$

3 If one of the class intervals for height is $10 < \text{height} \leq 19$, find the next equal-width class interval.

4 If the range of a set of continuous data is 28, find the size of each class interval if we group the data into:

a 4 equal-width class intervals.

b 7 equal-width class intervals.

5 Construct a grouped frequency table for the following data:

a $46,\ 54,\ 35,\ 23,\ 24,\ 28,\ 26,\ 11,\ 19,\ 17,\ 32$

b $83,\ 68,\ 39,\ 42,\ 86,\ 66,\ 64,\ 76,\ 63,\ 43,\ 65,\ 83,\ 63,$
$67,\ 49,\ 51,\ 32,\ 55,\ 38,\ 65,\ 41,\ 73,\ 35,\ 36,\ 74$

6 Consider the following set of scores:

$$30,\ 67,\ 24,\ 51,\ 49,\ 53,\ 36,\ 24,\ 57,\ 66,\ 26$$

a Determine a set of five class intervals that could be used to analyse this data.

b Construct a grouped frequency table for the data.

7 Consider the following set of scores:

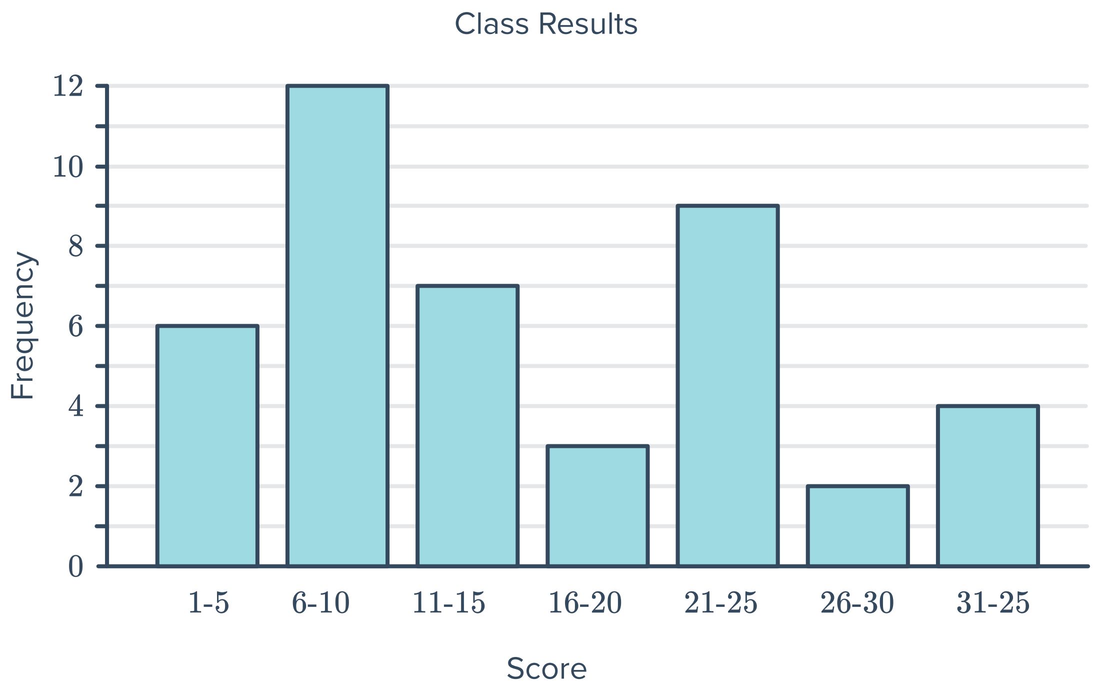
$$12,\ 59,\ 61,\ 27,\ 58,\ 18,\ 76,\ 27,\ 52,\ 19,\ 13,\ 56,\ 71,\ 31,\ 73,$$
$$60,\ 41,\ 17,\ 22,\ 68,\ 57,\ 15,\ 40,\ 19,\ 76,\ 44,\ 60,\ 55,\ 36$$

a Determine a set of seven class intervals that could be used to analyse this data.

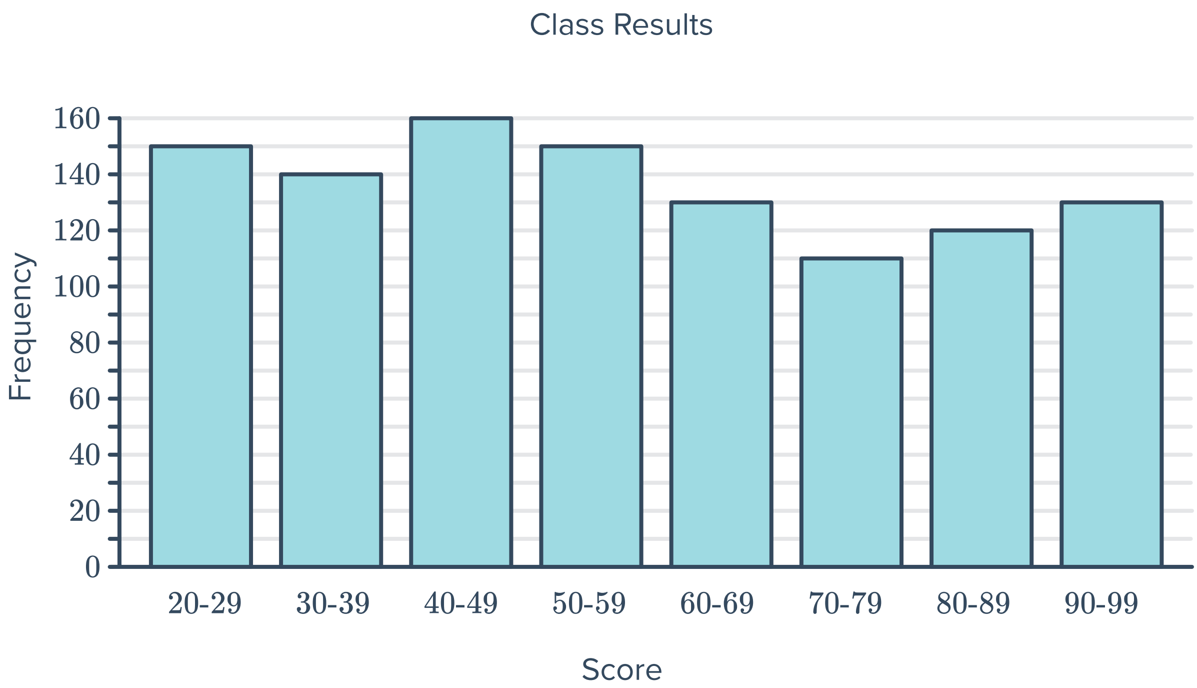
b Construct a grouped frequency table for the data.

8 Construct a frequency table for the follow[redacted]lumn graphs:

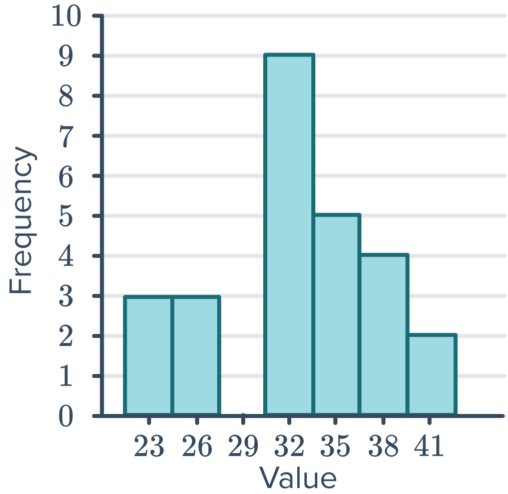
a



b



9 Consider the following frequency histogra



Class interval	Class centre	Frequency
$21.5 \leq p < 24.5$	23	
$24.5 \leq p < 27.5$	26	
$27.5 \leq p < 30.5$	29	
$30.5 \leq p < 33.5$	32	
$33.5 \leq p < 36.5$	35	
$36.5 \leq p < 39.5$	38	
$39.5 \leq p < 42.5$	41	

a Complete the grouped frequency table.

b Construct a frequency polygon from this data.

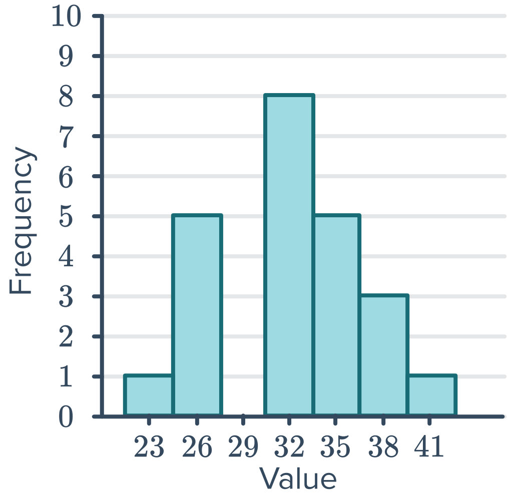
c How many values are in the class interval $27.5 \leq p < 30.5$?

d How many values fall between 24.5 inclusive, and 36.5?

e How many values are below 33.5?

f How many values are above and including 30.5?

10 Consider the following frequency histogram:



Class limits	Midpoint	Frequency
$21.5 - 24.5$	23	
$24.5 - 27.5$	26	
$27.5 - 30.5$	29	
$30.5 - 33.5$	32	
$33.5 - 36.5$	35	
$36.5 - 39.5$	38	
$39.5 - 42.5$	41	

a Complete the grouped frequency table.

b Construct a frequency polygon from this data.

c How many values are in the class 27.5 to 30.5?

d How many values fall between 24.5 and 36.5?

e How many values are below 33.5?

Applications

11 A survey was conducted which asked 30 people how many books they had read in the past month. Based on the frequency table provided, state whether the following statements are correct:

a 11 people have read from 6 to 10 books in the past month.

b 28 people have read at most 15 books in the past month.

c We cannot determine from the table how many people have read exactly 12 books.

d We can determine that 2 people have read exactly 5 books in the past month.

Number of books read	Frequency
$1-5$	2
$6-10$	11
$11-15$	15
$16-20$	2

12 The following frequency table below shows the resting heart rate of some people taking part in a study:

a Complete the table.

b How many people took part in the study?

c How many people had a resting heart rate between 55 and 74?

d How many people had a resting heart rate of below 65?

Heart Rate	Class Centre	Frequency
$45-54$		15
$55-64$		25
$65-74$		26
$75-84$		30

13 In a study of reaction times of dogs to a specific stimulus, an animal trainer obtained the following data, given in seconds:

a Construct a histogram and frequency polygon for this data.

b Compare the total area of the columns in the histogram and the area under the frequency polygon.

Reaction time t (seconds)	Frequency
$2.3 \leq t < 3.0$	14
$3.0 \leq t < 3.7$	10
$3.7 \leq t < 4.4$	6
$4.4 \leq t < 5.1$	9
$5.1 \leq t < 5.8$	3
$5.8 \leq t < 6.5$	1

14 In product testing, the number of faults detected in producing a certain machinery is recorded each day for several days. The frequency table shows the results:

a Construct a histogram to represent the data.

b What is the lowest possible number of faults that could have been recorded on any particular day?

Number of faults	Frequency
$0 - 3$	10
$4 - 7$	14
$8 - 11$	20
$12 - 15$	16

15 The following frequency table shows the average time spent travelling to work for 50 people:

Commute time (minutes)	Frequency
$0 \leq \text{time} < 20$	14
$20 \leq \text{time} < 40$	16
$40 \leq \text{time} < 60$	10
$60 \leq \text{time} < 80$	7
$80 \leq \text{time} < 100$	3
Total	50

a Construct a histogram to display the data shown in the frequency table.

b State whether the following statements about the data are accurate:

i The data shows that most people travel to work by car or by walking, since most travel times are fairly short, and only a few people travel by bus or train.

ii The data suggests that people prefer a shorter commute to work. A majority live within 40 minutes travel, and in general the longer the commute the less people there are in that category.

iii The data suggests that people don't care too much about how far away from work they live. Roughly equal portions of people live less than 40 minutes away and more than 40 minutes away.

iv The data shows that everyone lives within an hours travel from their work, with the peak amount of people living between 20 and 40 minutes away.

16 Infant mortality rate is measured as the nu[redacted] of child deaths per 1000 births. The infant mortality rate was measured in a number of countries and the results presented as a frequency histogram:

a State the following class intervals:

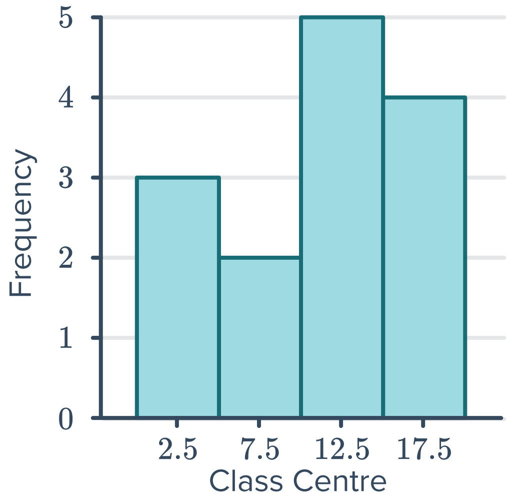
i First interval

ii Second interval

iii Third interval

iv Fourth interval

b How many of the countries recorded an infant mortality rate below 5?



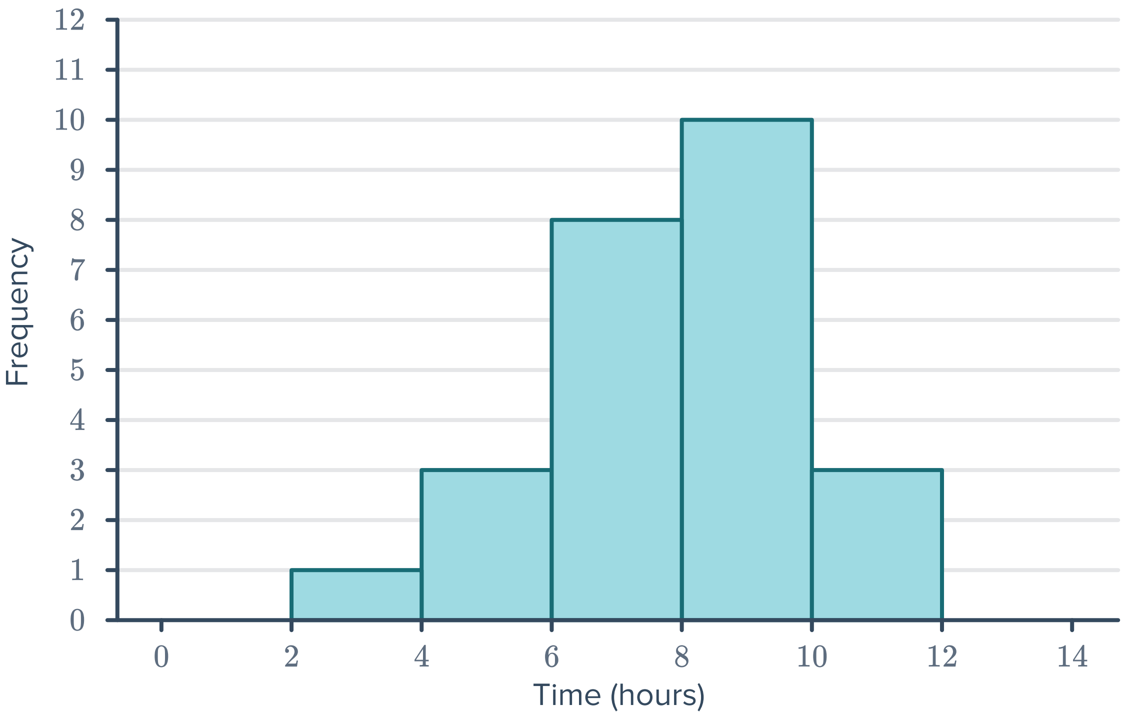
17 As part of a fuel watch initiative, the price of petrol, p, at a service station was recorded each day for 21 days. The frequency table shows the findings:

Price (in cents per litre)	Class Centre	Frequency
$120.9 < p \leq 125.9$	123.4	4
$125.9 < p \leq 130.9$	128.4	6
$130.9 < p \leq 135.9$	133.4	5
$135.9 < p \leq 140.9$	138.4	6

a Find the highest price that could have been recorded.

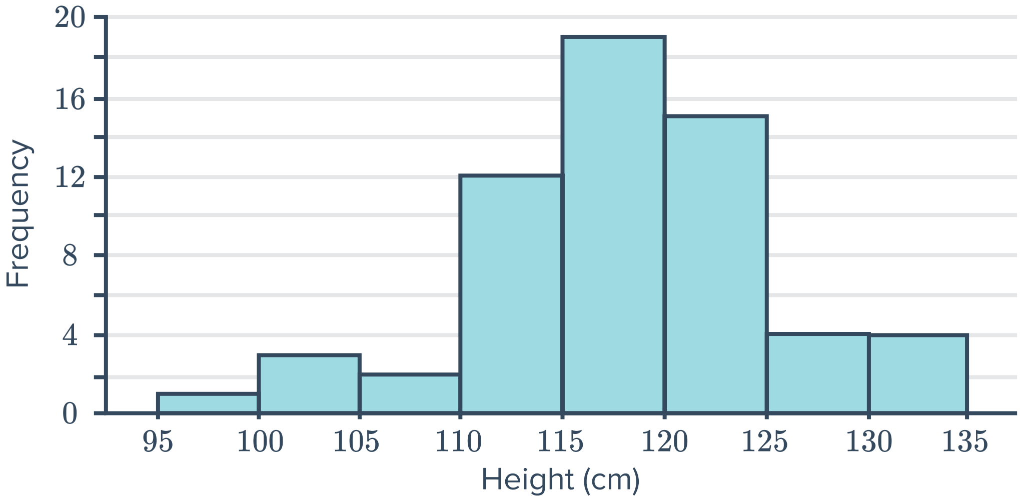
b How many days was the price above 130.9 cents?

18 The given histogram shows the number of [redacted]s that students in a particular class had slept for the night before:



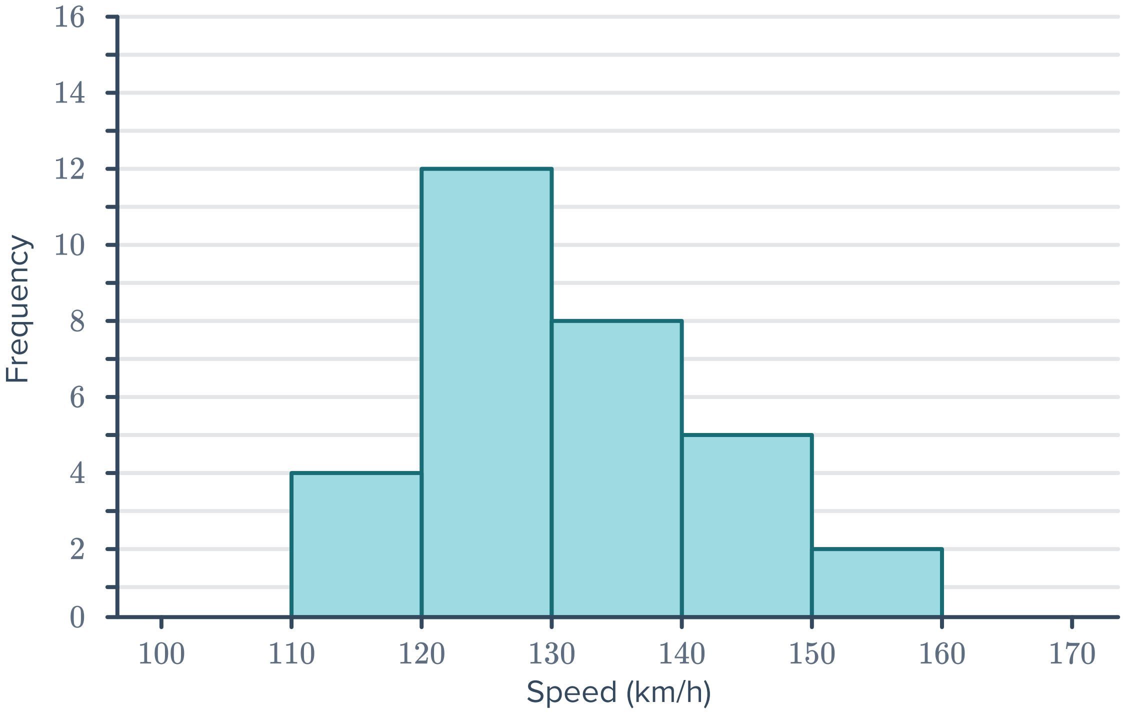
- **a** How many students are in the class?
- **b** How many students had at least 8 hours of sleep that night?
- **c** What percentage of students had less than 6 hours of sleep?

19 The histogram below shows the heights of 60 tomato plants growing in a greenhouse:



- **a** Create a frequency table for this set of data.
- **b** What percentage of the tomato plants are in the most common category? Round your answer to one decimal place.

20 The histogram below shows the speed of  delivered by cricket bowlers:



a Complete a frequency table above for this set of data.

b What percentage of balls were bowled at less than $130\ \mathrm{km/h}$? Round your answer to one decimal place.

21 The following frequency table shows the price of the most recent book that seventy two university students bought:

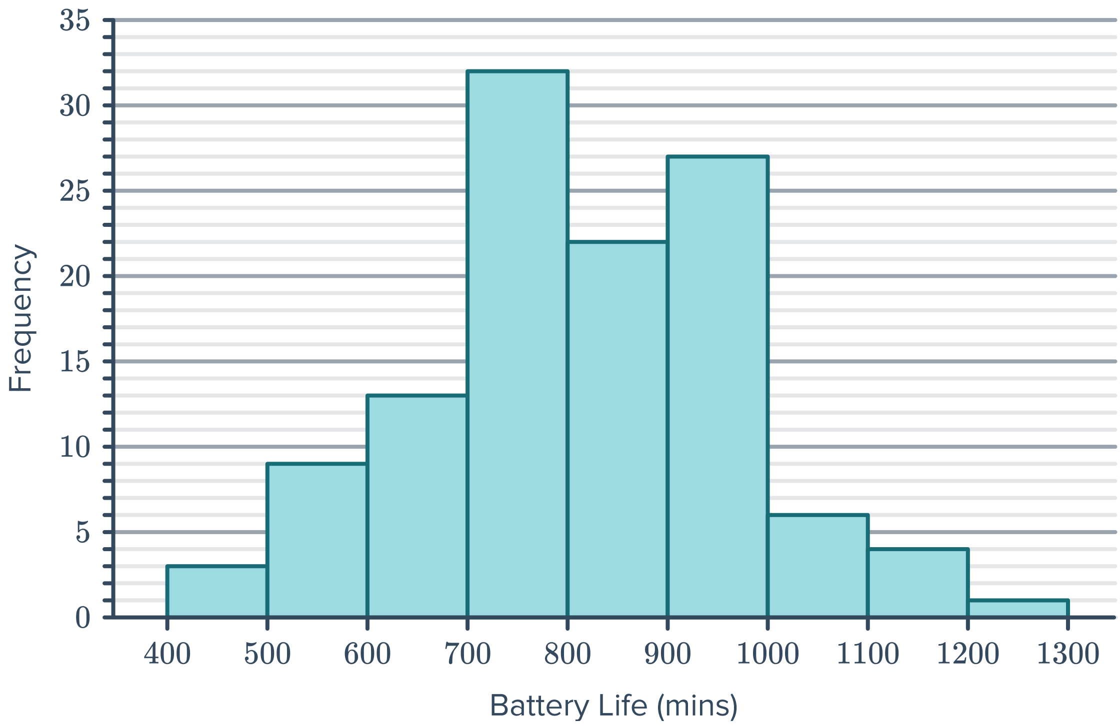
Price (\$)	Frequency
$0-19$	6
$20-39$	20
$40-59$	12
$60-79$	6
$80-99$	19
$100-119$	9
Total	72

a Construct a histogram to display the data shown in the frequency table.

b Which groups represent peaks in the data set?

c Describe what the data shows regarding the price university students generally pay for their books.

22 A battery manufacturer tests their batterie[illegible] portable music player, to find out how long the batteries last under constant use. The histogram below shows the data from their tests:



a Create a frequency table for this set of data.

b What percentage of batteries last longer than 10 hours? Round your answer to one decimal place.

c If the average length of a song is 4 minutes, what percentage of batteries will last for an average of 250 songs or more? Round your answer to one decimal place.

23 The following frequency table shows the data distribution for the length of leaves (in millimetres) collected from a species of tree in the botanical gardens:

a How many leaves less than 60 mm were collected?

b What was the most common interval of leaf length collected?

c Are leaves more likely to be at least 60 mm in length?

d Can we conclude from the table that there were no leaves collected with a length less than 5 mm? Explain your answer.

Leaf length	Frequency
$0 \leq x < 20$	5
$20 \leq x < 40$	11
$40 \leq x < 60$	19
$60 \leq x < 80$	49
$80 \leq x < 84$	43